



Title

rd2d — Location-based methods for boundary discontinuity designs.

Syntax

```
rd2d y x1 x2 assignment [if] [in], b(# # [# # ...]) [h(# | # # # # [# # # # ...])
fuzzy(varname) deriv(# [#]) tangvec(# # [# # ...]) p(#) q(#) kernel(kernel)
kerneltype(type) vce(vcetype) cluster(clustvar) level(#) side(side)
bwselect(selector) bwparam(target) method(method) bwcheck(#)
masspoints(masspointsoption) scaleregul(#) scalebiasrcrt(#) stdvars(on|off)
```

Description

rd2d estimates location-based local polynomial boundary discontinuity effects at one or more two-dimensional boundary evaluation points. The first three variables are the outcome and bivariate running variable; *assignment* is a binary treatment-side indicator.

The option **b**() supplies boundary points as pairs. For example, **b**(0 0 0 1) requests estimates at (0,0) and (0,1). The option **h**() accepts either a scalar bandwidth or four bandwidths per boundary point: control *x1*, control *x2*, treated *x1*, and treated *x2*.

Companion commands are **rdbw2d** for location-based bandwidth selection and **rd2d_dist** for distance-based designs. Requires Stata 16 or later.

Options

Boundary Points and Design

b(# # [# # ...]) specifies the boundary evaluation points and is required.

h() specifies user-provided bandwidths. A scalar uses the same bandwidth for all dimensions, groups, and evaluation points. Otherwise, supply four values for each evaluation point: *h01 h02 h11 h12*, corresponding to the first and second running-variable coordinates in the control group and the first and second coordinates in the treated group. If **h**() is omitted, **rd2d** computes automatic bandwidths using **rdbw2d**.

fuzzy(*varname*) specifies the treatment receipt/status variable for fuzzy boundary discontinuity designs. The main table reports the fuzzy Wald ratio. The reduced-form outcome discontinuity is stored in **e(itt)**, and the first-stage treatment receipt/status discontinuity is stored in **e(fs)**. Side-specific fuzzy tables are stored in **e(itt_0)**, **e(itt_1)**, **e(fs_0)**, and **e(fs_1)**.

deriv() specifies derivative orders in the first and second coordinates. For example, **deriv**(1 2) estimates the mixed derivative with one derivative in the first coordinate and two in the second coordinate. Default is **deriv**(0 0).

tangvec() supplies directional derivative vectors, one pair per boundary point. If supplied, **tangvec**() overrides **deriv**() and requires **p**(1) or higher.

Local Polynomial Regression

p() and **q**() set the local polynomial orders for point estimation and bias-corrected inference. Defaults are **p**(1) and **q**(**p**+1).

kernel() specifies the kernel function. Available choices are:
triangular or **tri**: triangular kernel, the default.
epanechnikov or **epa**: Epanechnikov kernel.
uniform or **uni**: uniform kernel.
gaussian or **gau**: Gaussian kernel.

kerneltype() specifies the kernel structure. Available choices are:
prod: product kernel, the default.
rad: radial kernel.

Variance-Covariance Estimation

vce() specifies the variance-covariance estimator. Available choices are:

- hc0**: heteroskedasticity-robust plug-in residual variance estimator without small-sample adjustment.
- hc1**: heteroskedasticity-robust plug-in residual variance estimator with HC1 degrees-of-freedom adjustment, the default.
- hc2**: heteroskedasticity-robust plug-in residual variance estimator with HC2 leverage adjustment.
- hc3**: heteroskedasticity-robust plug-in residual variance estimator with HC3 leverage adjustment.

cluster(*clustvar*) specifies a cluster ID variable. When supplied, cluster-level influence sums with degrees-of-freedom weights are used.

level() sets the confidence level. Default is **level(95)**.

side() specifies the type of confidence interval. Available choices are:

- two**: two-sided intervals, the default.
- left**: left-tail one-sided intervals.
- right**: right-tail one-sided intervals.

Bandwidth Selection

bwselect() specifies the automatic bandwidth selector. Available choices are:

- mserd**: one common MSE-optimal bandwidth selector for the boundary RD treatment-effect estimator at each evaluation point, the default.
- cerd**: CER-optimal counterpart of **mserd**.
- imserd**: IMSE-optimal common bandwidth selector for the boundary RD treatment-effect estimator based on all evaluation points.
- icerd**: CER-optimal counterpart of **imserd**.
- msetwo**: two different MSE-optimal bandwidth selectors, one for the control group and one for the treated group, at each evaluation point.
- certwo**: CER-optimal counterpart of **msetwo**.
- imsetwo**: two IMSE-optimal bandwidth selectors, one for each group, based on all evaluation points.
- icertwo**: CER-optimal counterpart of **imsetwo**.
- user provided**: used when **h()** is supplied.

bwparam() specifies the target parameter for fuzzy automatic bandwidth selection. Available choices are:

- main**: selects bandwidths for the linearized fuzzy Wald ratio, the default.
- itt**: selects bandwidths for the reduced-form outcome discontinuity. This option is ignored in sharp designs and when **h()** is supplied.

method() specifies the method for bandwidth calculations. Available choices are:

- dpi**: data-driven plug-in MSE-optimal selector, the default.
- rot**: rule-of-thumb selector.

bwcheck() enlarges preliminary bandwidths, when needed, so that at least the requested number of unique observations are included in each side-specific window. Default is **50 + p + 1**.

masspoints() specifies how repeated running-variable values are handled. Available choices are:

- check**: detects mass points and reports unique-observation counts, the default.
- adjust**: adjusts preliminary bandwidths to ensure a minimum number of unique observations within each side of the boundary.
- off**: ignores mass points.

scaleregul() specifies the scaling factor for the regularization term in bandwidth selection. Default is **scaleregul(3)** for **rd2d**.

scalebiasrcrt() specifies the scaling factor used for bias correction based on higher-order expansions. Default is **scalebiasrcrt(1)**.

stdvars() controls whether the running variables are standardized before automatic bandwidth selection. Use **stdvars(on)** or **stdvars(off)**. Default is **on**; this option affects automatic bandwidth selection only.

Examples

Simulated sharp design

```
. clear
. set obs 800
. set seed 123
. gen x1 = rnormal()
. gen x2 = rnormal()
. gen d = x1 >= 0
. gen y = 3 + 2*x1 + 1.5*x2 + d + rnormal()
. rd2d y x1 x2 d, b(0 0 0 1) h(.9) masspoints(off)
```

Fuzzy design

```
. gen takeup = runiform() < cond(d, .8, .2)
. gen yf = 3 + 2*x1 + 1.5*x2 + 1.5*takeup + rnormal()
. rd2d yf x1 x2 d, b(0 0 0 1) h(.9) fuzzy(takeup)
```

Stored results

`rd2d` stores the following in `e()`:

Matrices

<code>e(main)</code>	main treatment-effect table
<code>e(bw)</code>	bandwidth and effective sample-size table
<code>e(V_main)</code>	covariance matrix for the main bias-corrected estimates
<code>e(main_0)</code>	control-side table, sharp designs
<code>e(main_1)</code>	treated-side table, sharp designs
<code>e(itt)</code>	reduced-form table, fuzzy designs
<code>e(fs)</code>	first-stage table, fuzzy designs
<code>e(itt_0)</code>	control-side reduced-form table, fuzzy designs
<code>e(itt_1)</code>	treated-side reduced-form table, fuzzy designs
<code>e(fs_0)</code>	control-side first-stage table, fuzzy designs
<code>e(fs_1)</code>	treated-side first-stage table, fuzzy designs
<code>e(V_itt)</code>	covariance matrix for <code>e(itt)</code>
<code>e(V_fs)</code>	covariance matrix for <code>e(fs)</code>
<code>e(V_itt_0), e(V_itt_1)</code>	covariance matrices for side-specific reduced-form tables
<code>e(V_fs_0), e(V_fs_1)</code>	covariance matrices for side-specific first-stage tables

Scalars

<code>e(N)</code>	number of observations
<code>e(p)</code>	polynomial order for point estimation
<code>e(q)</code>	polynomial order for bias-corrected inference

References

- Cattaneo, M. D., Titiunik, R., and Yu, R. R. (2026). Estimation and Inference in Boundary Discontinuity Designs: Location-Based Methods.
- Cattaneo, M. D., Titiunik, R., and Yu, R. R. (2026). Boundary Discontinuity Designs: Theory and Practice.

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